

30/7/25

OHM'S LAW

The ratio of potential difference (V) b/w any two points on a conductor to the current (I) flowing b/w them, is constant, provide the temperature of conductor does not change.

$$R = \text{constant} \quad \left| \quad \frac{V}{I} = \text{constant}, \text{ or } V = IR \right.$$

R is the resistance of the conductor b/w the 2 points considered.

→ linear relationship b/w V & I doesn't apply to all non-metallic conductors.

example:- silicon carbide.

Relationship $V \propto I^m$ K & m are
doesn't apply to constant
all non-linear devices such
as Zener diode & voltage-

regulator (VR) tubes

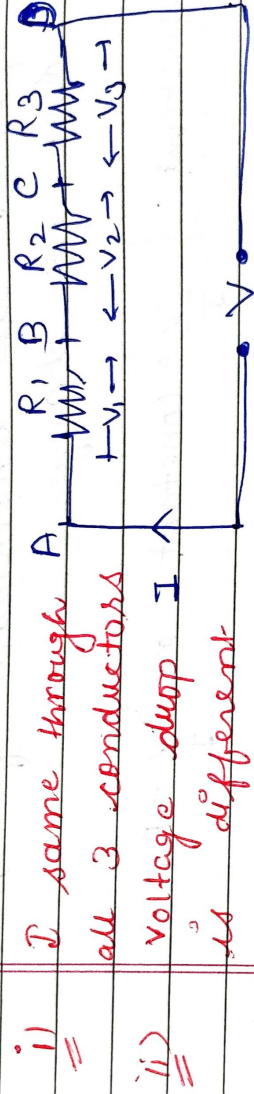
$$\frac{V}{I} = \text{constant} \Rightarrow V = IR$$

$$V = -IR$$

31/07/25 { 2/08/25 }

RESISTANCE IN SERIES

→ for example:- a resistance having 3 resistances R_1, R_2 & R_3 etc. & are connected in series.



i) I same through all 3 conductors
voltage drop is different across all due to diff resistance given by ohm's law

iii) sum of 3 voltage drop is equal to the voltage applied across the 3 conductors

$$I \times V = V$$

ohm's law.

$$V = V_1 + V_2 + V_3 = IR_1 + IR_2 + IR_3$$

$R =$ equivalent Resistance of series combination

$$IR = IR_1 + IR_2 + IR_3$$

or

$$R = R_1 + R_2 + R_3$$

also

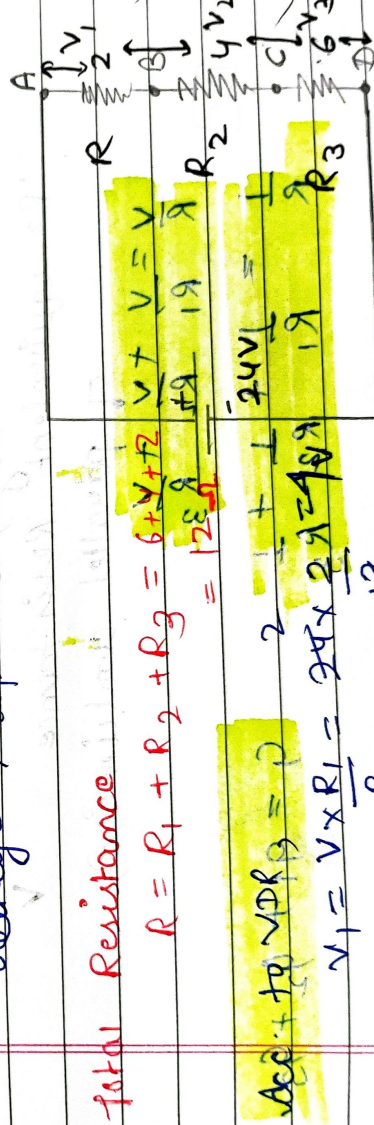
$$\frac{1}{G} = \frac{1}{G_1} + \frac{1}{G_2} + \frac{1}{G_3}$$

Main characters of series circuit

- 1) Same current to all part of c.c.
- 2) Diff. resistors have their individual voltage drop.
- 3) voltage drops are additive
- 4) applied voltage = sum of diff. voltage drop
- 5) Resistances are additive
- 6) Powers are additive

Voltage Divider Rule

→ In a series circuit, same current flows through each of the given resistors, voltage drop varies with resistor

Total Resistance

$$R = R_1 + R_2 + R_3 = 6 + 4 + 2 = 12 \Omega$$

$$V_1 = V \times \frac{R_1}{R}$$

$$V_1 = 24 \times \frac{6}{12} = 12V$$

$$V_2 = V \times \frac{R_2}{R} = 24 \times \frac{4}{12} = 8V$$

$$V_3 = V \times \frac{R_3}{R} = 24 \times \frac{2}{12} = 4V$$

RESISTANCE IN PARALLEL

Potential difference across all resistances is same.

Current in each resistor is diff & given by Ohm's law.

The total current is sum of 3 separate currents.

$$I = I_1 + I_2 + I_3$$

$$= \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$I = \frac{V}{R} \quad (V = \text{applied voltage})$$

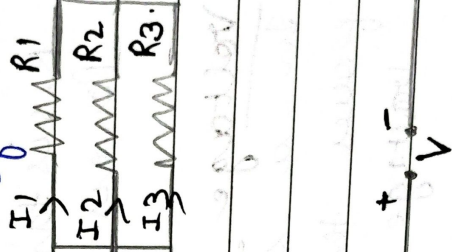
(R = equivalent Resistance)

or parallel combination)

$$\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$G = G_1 + G_2 + G_3$$

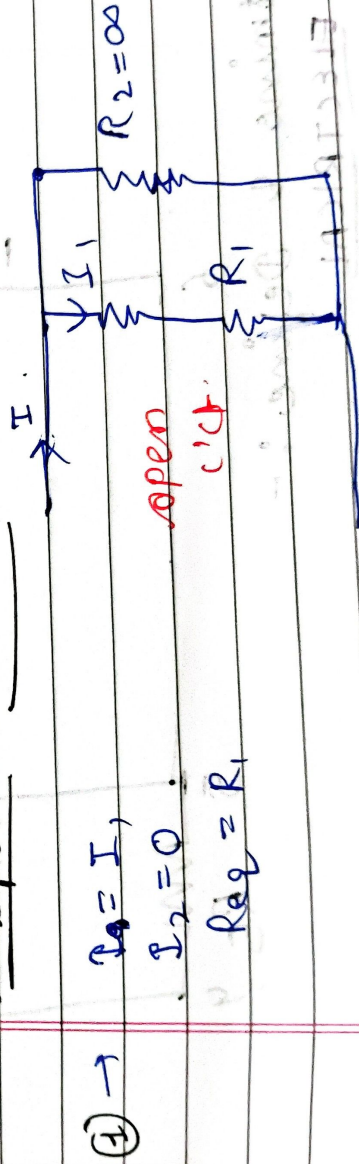


MAIN CHARACTERISTICS OF PARALLEL

- Same voltage acts across all parts of circuit
- different resistors have their diff individual current
- branch currents are additive
- Conductances are additive
- powers are additive.

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} \quad \rightarrow \text{for 2 Resistors.}$$

2 special Cases:



②

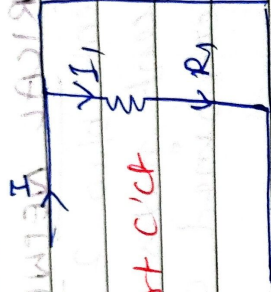
$$I_1 R_2 = 0$$

$$I_1 = 0$$

$$I_2 = I$$

$$R_{eq} = R$$

Short C'kt

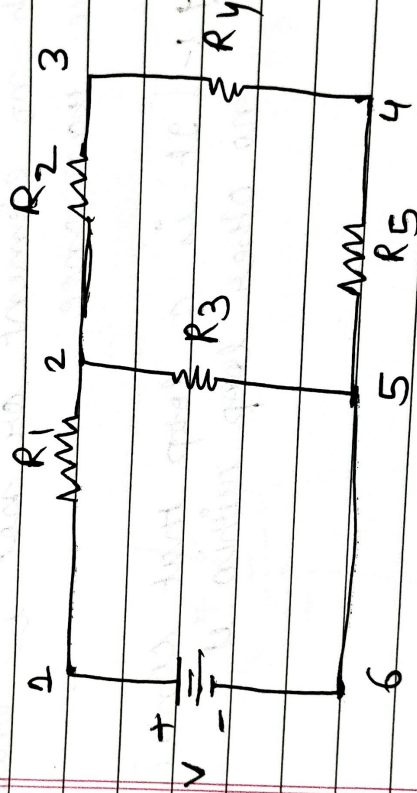


$$I_2 = I$$

$$V = I_2 R_2 = 0$$

$$V = I_1 R_1$$

$$V = 0$$



1) Count & Define:-

① circuit.

is a complete circular path in which electrical current flows.

② Electrical Network:-

A collection of interconnected electrical components like resistors, capacitors, inductors & voltage sources.

③ Node:-

A point in a c't where 2 or more c't containing elements are connected together.

If there is no element b/w 2 or more connected adjacent nodes then these nodes are combined as a single node.

→ 5 nodes

④ Branch:-

That part of network connected b/w 2 nodes

→ 6 branches

⑤ loop:-

Closed path in a c't in which no element or node is encountered more than once.

→ 3 loops

⑥

Mesh.

It is a loop that contains no other loop within it.

•) 3 meshes

#

Kirchoff's Point Law or Current Law
(KCL)

Leaving = -ve

Incoming = +ve.

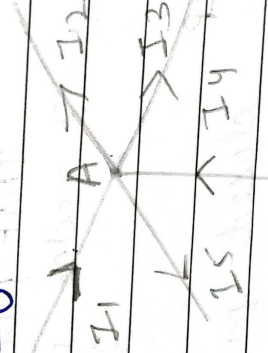
In any electrical network, the algebraic sum of the currents meeting at a point (or junction) is zero.

Current leaving = Current entering.
→ No accumulation of charge at the junction of the network.

$$eq^n = I_1 - I_2 - I_3 + I_4 - I_5 = 0$$

$$I_1 + I_4 = I_2 + I_3 + I_5$$

$$\sum I = 0$$



Kirchoff's Mesh law or Voltage law (KVL)

The algebraic sum of the products of current & resistances in each of the conductors in any closed path (or mesh) in a network plus the algebraic sum of the emf's in that path is zero.

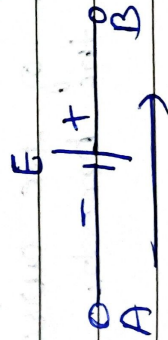
$$\Sigma IR + \Sigma e.m.f = 0$$

NOTE:- Algebraic sum is the sum which takes into account the polarities of the voltage drop.

SIGN CONVENTION OF VOLTAGE SIGN(a) Sign of battery E.M.F.

Terminal
-ve to +ve = +ve sign
+ve to -ve = -ve sign

Rise in voltage = +ve.
fall in voltage = -ve.



Rise in voltage

$$+E$$



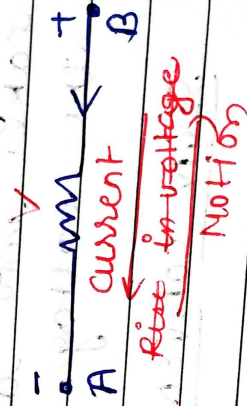
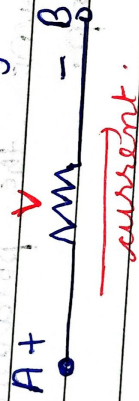
fall in voltage

$$-E$$

9a) Sign of IR Drop:

→ Same direction as the current
= fall in potential
(because current flows from a
higher to a lower potential)
{ voltage = -ve }

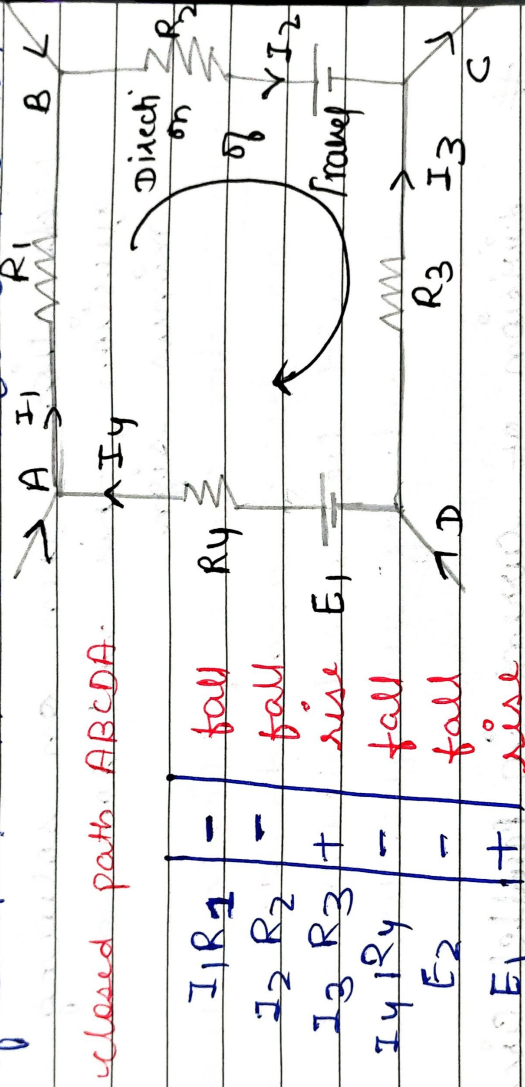
→ If we go in a direction opposite
to that of the current =
rise in voltage
{ voltage = +ve }



Rise in voltage
[+V = +IR]

→ Sign of battery e.m.f is independent
of the direction of the current
through the branch.

→ Sign of voltage drop across a resistor depends on the direction of current through that resistor but is independent of the polarity of any other source of e.m.f. in the ckt under consideration.



Using KVL,

$$-I_1 R_1 - I_2 R_2 + I_3 R_3 - I_4 R_4 - E_2 + E_1 = 0$$

$$I_3 R_3 + E_1 = I_1 R_1 + I_2 R_2 + E_2$$

$$I_4 R_4 + I_1 R_1 + I_2 R_2 - I_3 R_3 = E_1 - E_2$$

Assumed Direction of Current

- assume clockwise or anticlockwise
- If the assumed direction of current is not the actual direction, then on solving question, this will have a minus sign

Important:- ~~the~~ Once a particular direction has been assumed, the same should be used throughout the sol. of the question

Division of Current in Parallel Circuits Current Divider Rule

$$I_1 = \frac{V}{R_1}$$

$$I = \frac{I_{\text{Req}}}{R_1}$$

$$= \frac{I R_1 R_2}{R_1 (R_1 + R_2)}$$

$$I_1 = \frac{I R_2}{R_1 + R_2}$$

+ve -



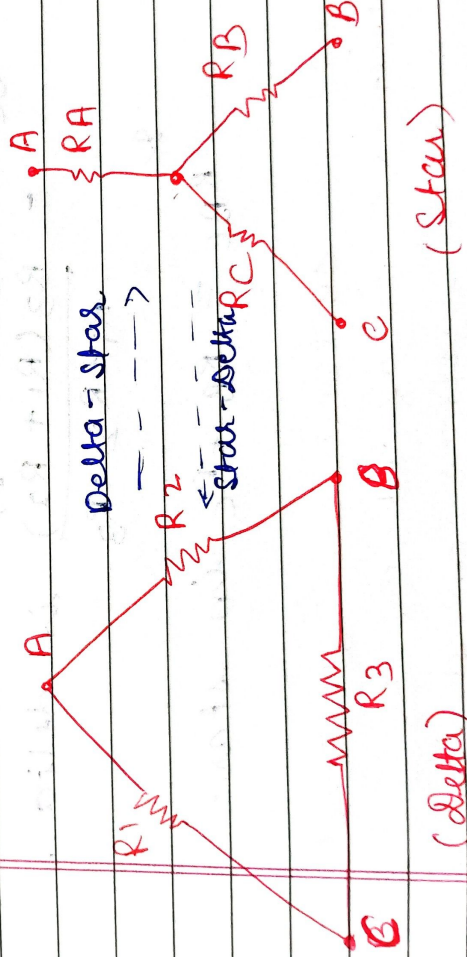
Transformation of Delta-star or vice versa.

1) Delta.

3 resistances connected to nose to tail are said to be delta (Δ) or mesh connected (they form a mesh)

2) Star.

3 resistances connected together at a common point O are said to be star (or Y) connected



* Delta-Star Transformation

The replacement of delta or mesh by equivalent star system is known as delta-star transformation.

→ The 2 systems will be equivalent or identical if the resistances measured b/w any pair of line is open.

Resistances b/w B & C

$$\begin{aligned} R_{BC} &= R_A \parallel (R_1 + R_2) \\ &= \frac{R_3(R_1 + R_2)}{R_1 + R_2 + R_3} \end{aligned} \quad \text{in delta,}$$

2

$$R_{BC} = R_B + R_C \text{ in star.}$$

Since 2 systems are identical resistances measured b/w B & C in both systems must be equal.

$$\textcircled{1} \quad R_B + R_C = \frac{R_3(R_1 + R_2)}{R_1 + R_2 + R_3}$$

Similarly solve c And A.

$$R_c + R_A = \frac{R_1(R_2 + R_3)}{R_1 + R_2 + R_3} \quad \text{--- (2)}$$

11'ly, solve A & B.

$$R_A + R_B = \frac{R_2(R_1 + R_3)}{R_1 + R_2 + R_3} \quad \text{--- (3)}$$

By adding eqⁿ (1) (2) & (3)

$$2(R_A + R_B + R_C) = \frac{2(R_1 R_3 + R_2 R_3 + R_1 R_2)}{R_1 + R_2 + R_3}.$$

or

$$R_A + R_B + R_C = \frac{R_1 R_3 + R_2 R_3 + R_1 R_2}{R_1 + R_2 + R_3}. \quad \text{--- (4)}$$

By subtracting eqⁿ 1, 2, 3 from 4.

$$R_A = \frac{R_1 R_2}{R_1 + R_2 + R_3} \quad \text{--- (5)}$$

$$R_B = \frac{R_1 R_3}{R_1 + R_2 + R_3} \quad \text{--- (6)}$$

$$R_C = \frac{R_2 R_3}{R_1 + R_2 + R_3} \quad \text{--- (7)}$$

Relationship can be expressed as:-

The equivalent star resistance connected to a given terminal is equal to the product of 2 delta resistances connected to the same terminal divided by the sum of delta connected resistances.

If the 3 delta - connected resistances have the same value R_D . The 3 resistances in the equivalent star for identical systems will be

$$\textcircled{8} \quad R_S = \frac{R_D R_D}{R_D + R_D + R_D} = \frac{R_D}{3}$$

Star-Delta Transformation

Multiply eqn (5) & (6), (6) & (7), (7) & (8)
 & then adding we get

$$R_A R_B + R_B R_C + R_C R_A = \frac{R_1 R_2^2 R_3 + R_1 R_3^2 R_2 + R_1^2 R_2 R_3}{(R_1 + R_2 + R_3)^2}$$

$$(9) \quad \frac{R_1 R_2 R_3 (R_2 + R_3 + R_1)}{(R_1 + R_2 + R_3)^2} = \frac{R_1 R_2 R_3}{R_1 + R_2 + R_3}$$

Dividing eqn (9) by eqn (5), (6) & (7)
 seperately we get

$$R_3 = \frac{R_A R_B + R_B R_C + R_C R_A}{R_A} = R_B + R_C + \frac{R_B R_C}{R_A} \quad (10)$$

$$R_1 = \frac{R_A R_B + R_B R_C + R_C R_A}{R_B} = R_A + R_C + \frac{R_A R_C}{R_B} \quad (11)$$

2

$$R_2 = \frac{R_A R_B + R_B R_C + R_C R_A}{R_C} = R_A + R_B + \frac{R_A R_B}{R_C} \quad (12)$$

Relationship may be expressed as:-

The equivalent delta resistance b/w 2 terminals is the sum of 2 star resistances connected to those terminals plus the product of the same divided by 3 star resistance.

If the 3 star^{connected} resistances have the same value R_s , the 3 Δ resistances of the equivalent delta for identical systems will be

$$R_D = R_s + R_s + \frac{R_s R_s}{R_s} = 3 R_s$$